

Classwork Problems

Unit 3: Integration

L22-24: Integration – Introduction to integration and integrals of standard functions

Basic formulae problems:

Evaluate the following:

1. $\int (5x^2 - 8x + 5) dx$

Ans. $\frac{5x^3}{3} - 4x^2 + 5x + C$

2. $\int \left(\sqrt{x} + \frac{1}{3\sqrt{x}} \right) dx$

Ans. $\frac{2}{3}x^{\frac{3}{2}} + \frac{2}{3}x^{\frac{1}{2}} + C$

3. $\int 4e^{-7x} dx$

Ans. $-\frac{4}{7}e^{-7x} + C$

4. $\int \left(\frac{8}{x} - \frac{5}{x^2} + \frac{6}{x^3} \right) dx$

Ans. $8 \log x + \frac{5}{x} - \frac{3}{x^2} + C$

5. $\int (6^x + x^2 + e^x) dx$

Ans. $\frac{6^x}{\log 6} + 2x + e^x + C$

6. $\int \left(1 + \frac{1}{x} \right) dx$

Ans. $x + \log x + c$

7. $\int \frac{1}{x^5} dx$

Ans. $\frac{x^{-4}}{-4} + c$

8. $\int 5^x + 4e^{5x} dx$

Ans. $\frac{5^x}{\log 5} + \frac{4}{5}e^{5x} + c$

9. $\int \frac{1}{4+x^2} dx$

Ans. $\frac{1}{2} \tan^{-1} \frac{x}{2} + c$

10. $\int x^{\frac{2}{3}} + \frac{1}{x} + 5 dx$

Ans. $\frac{3x^{\frac{5}{3}}}{5} + \log x + 5x + c$

L25-26: Integration of Trigonometric functions

Evaluate the following:

1. $\int (\sin x + 8 \cos x) dx$

Ans. $-\cos x + 8 \sin x + C$

2. $\int 4 \sin \left(\frac{\theta}{3} \right) d\theta$

Ans. $-12 \cos\left(\frac{\theta}{3}\right) + C$

3. $\int \sec x (\sec x + \tan x) dx$

Ans. $\tan x + \sec x + C$

4. $\int \frac{\sin x + \cos x}{\cos x} dx$

Ans. $\log(\sec x) + x + C$

5. $\int \frac{1 - \sin x}{\cos x} dx$

Ans. $\log(\sec x + \tan x) - \log(\sec x) + C$

6. $\int \left(x + \frac{1}{\sqrt{1-x^2}}\right) dx$

Ans. $\frac{x^2}{2} + \sin^{-1}x + C$

7. $\int \frac{1}{4+x^2} dx$

Ans. $\frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + C$

8. $\int \frac{1}{x\sqrt{x^2-9}} dx$

Ans. $\frac{1}{3} \sec^{-1}\left(\frac{x}{3}\right) + C$

9. $\int \cos^2 x dx$

Ans. $\frac{x}{2} + \frac{1}{4} \sin 2x + C$

10. $\int \sin 2x \cdot \cos 3x dx$

Ans. $\frac{-1}{10} \cos 5x + \frac{1}{2} \cos x + C$

L27-29: Integration by Parts (Using LIATE rule)

Evaluate the following:

1. $\int x \cdot \sin x dx$

Ans. $-x \cdot \cos x + \sin x + C$

2. $\int x \cdot e^x dx$

Ans. $x \cdot e^x - e^x + C$

3. $\int \log x dx$

Ans. $x \cdot \log x - x + C$

4. $\int e^x \cdot \sin x dx$

Ans. $\frac{e^x}{2} (\sin x - \cos x) + C$

5. $\int x \cdot \log 2x dx$

Ans. $\frac{x^2}{2} \cdot \log 2x - \frac{x^2}{8} + C$

6. $\int x \cdot \sin 2x dx$

7. $\int x^2 \cdot \log x dx$

8. $\int x^2 \cdot e^x dx$

9. $\int \tan^{-1} x dx$

10. $\int (x^2 + 1) \cdot \log x \, dx$

Additional Problems

Evaluate the following integrals

1. $\int x^7 dx$

2. $\int \frac{1}{\sqrt{x}} dx$

3. $\int \sqrt[3]{x} \, dx$

4. $\int \frac{\cos x}{\sin^2 x} dx$

5. $\int \frac{dx}{\sin^2 x}$

6. $\int 4^x dx$

7. $\int \frac{x^2+1}{x^3} dx$

8. $\int \left(\frac{4e^{5x} - 9e^{4x} - 3}{e^{3x}} \right) dx$

9. $\int \sec^2(2 + 3x) dx$

10. $\int \sec(3x + 5) \tan(3x + 5) dx$

11. $\int (x + 1)^4 dx$

12. $\int \frac{dx}{\sqrt{5-9x}}$

13. $\int e^{3-8x} dx$

14. $\int \cos 4x \cos 2x \, dx$

15. $\int x^2 \sin x dx$

16. $\int \sin^{-1} x dx$

17. $\int x \cos 2x dx$

18. $\int x^2 e^{3x} dx$

19. $\int x (\log x)^2 dx$